

Manohhar Swarna

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Professional Summary

AI/ML Engineer with 4+ years of experience building and deploying AI-driven solutions across Machine Learning, Deep Learning, NLP, and Generative AI. Skilled in the end-to-end ML lifecycle—from data preprocessing and model development to deployment and MLOps. Experienced in building AI agents, fine-tuning LLMs (LoRA/QLoRA), and implementing RAG-based systems using CrewAI, LangChain, and Hugging Face. Proficient with AWS/GCP, Docker, CI/CD, and collaborative delivery in production environments.

Technical Skills

- **Programming & ML:** Python, SQL, XGBoost, Random Forest, Feature Engineering, Model Evaluation & Tuning
- **Generative AI & NLP:** RAG Pipelines, LLM Fine-tuning (LoRA/QLoRA), Prompt Engineering, Transformer Models (BERT, GPT), Embeddings, Semantic Search
- **Frameworks & Tools:** Hugging Face, PyTorch/TensorFlow, Scikit-learn, LangChain, CrewAI, FastAPI/Flask
- **Cloud & MLOps:** AWS (SageMaker, Lambda, EC2), GCP, Docker, Kubernetes, CI/CD, Model Deployment & Monitoring
- **Databases & Delivery:** SQL, Vector DBs (Pinecone, FAISS, Chroma), Git, Agile/Scrum, Jira, Stakeholder Management

Professional Experience

AI/Machine Learning Engineer, *Voiceflow*

Oct 2024 – Present | Remote, USA

- Worked on assigned tickets for a Tier-1 AI customer support Conversation Orchestrator using LangChain, FastAPI, and Redis, improving conversation flow handling and session continuity.
- Assisted in maintaining Redis-based session context (read/write, expiry, caching) to support smoother multi-turn conversations across agent runtime.
- Supported enhancements in the RAG Retrieval Service, including retrieval logic updates such as chunking adjustments and search tuning using Pinecone.
- Contributed to implementing basic metadata filtering in retrieval queries to ensure answers were generated from correct knowledge sources.
- Helped enable citation/source attachment in RAG outputs using LangChain and vector search results, improving response traceability for internal QA.
- Participated in production support (debugging logs, handling API errors/timeouts, regression validation) in a microservices environment.

Machine Learning Engineer, *Applied Materials*

Oct 2019 – Jun 2022 | India

- Designed and optimized NLP preprocessing pipelines (tokenization, cleaning, feature extraction) using Python, improving input quality and downstream model performance
- Built and maintained text embedding pipelines for training and inference, enhancing model accuracy and reducing noise in predictions
- Automated data migration and batch processing workflows using Python, enabling reliable, repeatable data transfers across environments and reducing manual effort
- Developed monitoring and troubleshooting utilities by analysing logs and system metrics (memory usage, JVM pauses), improving production stability and reducing issue resolution time.

Project

Agentic Banking AI Chatbot

[Github](#)

Built an agent-based banking chatbot using LangChain and GPT-4o, where the model dynamically selects and invokes tools (SQLite database, external APIs, web search) based on user queries. Implemented stateful multi-turn conversations by storing interactions in SQLite, and designed modular components for chat orchestration, tool execution, and schema validation. Added structured logging and custom exception handling to improve reliability, traceability, and debugging.

MCP-based Agentic Banking Chatbot

[Github](#)

Built an MCP-based agentic banking chatbot using LangChain and GPT-4o, where tools are decoupled and exposed via a Model Context Protocol (MCP) server over HTTP, enabling dynamic tool discovery through an MCP client instead of

hardcoded integrations. Implemented an async architecture (async/await) with aiosqlite for stateful multi-turn memory, and designed a modular client-server system with distributed tool execution (DB, CSV, APIs, web search). Added structured logging and custom exception handling to improve reliability, traceability, and performance, making the system scalable and production-ready.

RAG-based Resume Assistant

[Github](#)

Built a Retrieval-Augmented Generation (RAG) chatbot using LangChain and ChromaDB to answer queries grounded in resume data. Implemented an end-to-end pipeline including PDF parsing, text chunking, embedding generation, and similarity-based Top-K retrieval, and used retrieved context to construct prompts for accurate LLM responses. Designed the system with persistent vector storage, modular architecture, structured logging, and exception handling to ensure maintainability and consistent performance.

Education

Master of Science , <i>University of Central Missouri</i> Computer Science	Aug 2022 – Dec 2023 Missouri, USA
Bachelor of Technology , <i>Acharya Nagarjuna University</i> Computer Science	Jun 2015 – May 2019 Andhra Pradesh, India